

FULL DETAILS (Read-only) -> [Click Here to Create PDF for Current Dataset of Trial](#)

CTRI Number	CTRI/2024/01/061396 [Registered on: 12/01/2024] Trial Registered Prospectively															
Last Modified On:	15/10/2024															
Post Graduate Thesis	No															
Type of Trial	Observational															
Type of Study	Case Control Study															
Study Design	Other															
Public Title of Study	Study of Pre-identified methylation marker gene as a gastric cancer biomarker															
Scientific Title of Study	Validation and establishment of pre-identified novel gene(s) as gastric cancer biomarker(s)															
Trial Acronym	NIL															
Secondary IDs if Any	<table><tr><th>Secondary ID</th><th>Identifier</th></tr><tr><td>NIL</td><td>NIL</td></tr></table>	Secondary ID	Identifier	NIL	NIL											
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Details of Principal Investigator or overall Trial Coordinator (multi-center study)	<table><tr><td>Name</td><td>Srimoyee Ghosh</td></tr><tr><td>Designation</td><td>Professor</td></tr><tr><td>Affiliation</td><td>North Eastern Hill University</td></tr><tr><td>Address</td><td>Dept of Zoology,North Eastern Hill University Shillong North Eastern Hill University East Khasi Hills MEGHALAYA 793022 India</td></tr><tr><td>Phone</td><td>09774566179</td></tr><tr><td>Fax</td><td></td></tr><tr><td>Email</td><td>srimoyee2009@gmail.com</td></tr></table>		Name	Srimoyee Ghosh	Designation	Professor	Affiliation	North Eastern Hill University	Address	Dept of Zoology,North Eastern Hill University Shillong North Eastern Hill University East Khasi Hills MEGHALAYA 793022 India	Phone	09774566179	Fax		Email	srimoyee2009@gmail.com
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Source of Monetary or Material Support	Department of Zoology, North-Eastern Hill University, Shillong, Meghalaya-793022																
Primary Sponsor	<table border="1"> <tr> <td>Name</td><td>North Eastern Hill University</td></tr> <tr> <td>Address</td><td>Umshing, Mawkynroh, Shillong-793022, Meghalaya,India</td></tr> <tr> <td>Type of Sponsor</td><td>Other [Intramural funding]</td></tr> </table>	Name	North Eastern Hill University	Address	Umshing, Mawkynroh, Shillong-793022, Meghalaya,India	Type of Sponsor	Other [Intramural funding]										
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Countries of Recruitment	India																
Sites of Study Modification(s)	<table border="1"> <tr> <th colspan="4">No of Sites = 2</th></tr> <tr> <th>Name of Principal Investigator</th><th>Name of Site</th><th>Site Address</th><th>Phone/Fax/Email</th></tr> <tr> <td>Dr Gaurav Das</td><td>Dr. B. BOROOAH CANCER INSTITUTE</td><td>Department of Surgical Oncology, Dr. B. BOROOAH CANCER INSTITUTE,Gopinath Nagar, A K Azad Road, Guwahati-781016 Kamrup ASSAM</td><td>08638149432 das.drgaurav@gmail.com</td></tr> <tr> <td>Prof Srimoyee Ghosh</td><td>North East Cancer Hospital and Research Institute</td><td>Department of Biochemistry, North East Cancer Hospital and Research Institute, 11th Mile Amerigog, Jorabat Guwahati-781023 Assam, India Kamrup ASSAM</td><td>09774566179 srimoyee2009@gmail.com</td></tr> </table>	No of Sites = 2				Name of Principal Investigator	Name of Site	Site Address	Phone/Fax/Email	Dr Gaurav Das	Dr. B. BOROOAH CANCER INSTITUTE	Department of Surgical Oncology, Dr. B. BOROOAH CANCER INSTITUTE,Gopinath Nagar, A K Azad Road, Guwahati-781016 Kamrup ASSAM	08638149432 das.drgaurav@gmail.com	Prof Srimoyee Ghosh	North East Cancer Hospital and Research Institute	Department of Biochemistry, North East Cancer Hospital and Research Institute, 11th Mile Amerigog, Jorabat Guwahati-781023 Assam, India Kamrup ASSAM	09774566179 srimoyee2009@gmail.com
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Regulatory Clearance Status from DCGI	<table border="1"> <tr> <td>Status</td></tr> <tr> <td>Not Applicable</td></tr> </table>	Status	Not Applicable														
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Health Condition / Problems Studied	Health Type	Condition	
	Patients	(1) ICD-10 Condition: C169 Malignant neoplasm of stomach, unspecified,	
Intervention / Comparator Agent	Type	Name	Details
	Intervention	NIL	NIL
	Comparator Agent	NIL	NIL
Inclusion Criteria	Age From	18.00 Year(s)	
	Age To	99.00 Year(s)	
	Gender	Both	
	Details	1. Gastric cancer patients attending the surgical oncology OPD. 2. Patients who are willing to participate. 3. Age ≥ 18.	
ExclusionCriteria	Details	1. History of combination of other types of cancer. 2. Non-availability of matched cancer and normal tissue from the same patient. 3. Non-availability of patient data.	
Method of Generating Random Sequence	Not Applicable		
Method of Concealment	Not Applicable		
Blinding/Masking	Open Label		
Primary Outcome	Outcome	TimePoints	
	1. Degree of methylation of the proposed genes. 2. Expressional difference of the proposed genes.	1.1st week of participation:Gene expression analysis. 2.2nd week of participation: Gene methylation analysis.	
Secondary Outcome	Outcome	TimePoints	
	Association between the specific gene methylation with TNM status, Personal habits.	3rd week of participation.	
Target Sample Size	Total Sample Size="76" Sample Size from India="76" Final Enrollment numbers achieved (Total)= "Applicable only for Completed/Terminated trials" Final Enrollment numbers achieved (India)="Applicable only for Completed/Terminated trials"		
Phase of Trial	N/A		

Date of First Enrollment (India)	21/01/2024
Date of Study Completion (India)	Applicable only for Completed/Terminated trials
Date of First Enrollment (Global)	Date Missing
Date of Study Completion (Global)	Applicable only for Completed/Terminated trials
Estimated Duration of Trial	Years="3" Months="0" Days="0"
Recruitment Status of Trial (Global) Modification(s)	Not Applicable
Recruitment Status of Trial (India)	Open to Recruitment
Publication Details	N/A
Individual Participant Data (IPD) Sharing Statement	Will individual participant data (IPD) be shared publicly (including data dictionaries)? Response - YES - What data in particular will be shared? Response - Individual participant data that underlie the results reported in this article, after de-identification (text, tables, figures, and appendices). - What additional supporting information will be shared? Response - Study Protocol - Who will be able to view these files? Response - Researchers whose proposed use of the data has been approved by an independent review committee identified for this purpose. - For what types of analyses will this data be available? Response - To achieve aims in the approved proposal. - By what mechanism will data be made available? Response - Proposals should be directed to [srimoyee2009@gmail.com]. - For how long will this data be available <i>start date provided 31-12-2023 and end date provided 28-11-2023</i>? Response - Beginning 9 months and ending 36 months following article publication. - Any URL or additional information regarding plan/policy for sharing IPD? Additional Information - NIL
Brief Summary	INTRODUCTION <i>Epigenetic changes are linked with many cancer types [1]. In GC, Several genes were found to be differentially methylated compared to their</i>

adjacent normal cells of the same individual [2], and these differentially methylated genes can be used as a biomarker for GC screening, treatment, and prognosis [3].

In India, GC is responsible for about 53,253 deaths in the year 2020 and is the 6th leading cause of death [4]. 1 out of 160 men and 319 women has the risk to develop GC in their lifetime (0-70 years) [5]. Aizawl and Papumpare district of Mizoram and Arunachal Pradesh respectively has the highest incidence of GC among men and women respectively [6]. Kamrup (metro) of Assam has reported having the highest increase in APC (annual percentage change) of 6.5 in males and 11.7 in females [5].

Five Novel hypermethylated genes (PPP1R16B, FGF12, FEZF2, GDF7, and KCNQ5) were identified in the ethnic population of Mizoram from our laboratory at NEHU [7]. The primary objective of this present study is to establish these 5 genes as biomarker(s) of GC in patients from different ethnicities of Northeast India. These biomarker(s) will help clinicians better understand the disease, diagnose early and design the treatment plan accordingly.

HYPOTHESIS

The hypothesis of the study is PPP1R16B, FGF12, FEZF2, GDF7, and KCNQ5 genes are differentially methylated and have expressional differences between cancer matched-control tissues suggesting their potential utility as biomarkers for the diagnosis and prognosis of gastric cancer for different population.

OBJECTIVES

- a. To study the degree of methylation of PPP1R16B, FGF12, FEZF2, GDF7, and KCNQ5 in cancer and matched control tissue and measure its association with different patient parameters.
- b. To study the expression of PPP1R16B, FGF12, FEZF2, GDF7, and KCNQ5 in cancer and matched control tissue and measure its association with different patient parameters.
- c. To calculate the association between methylation and expression of PPP1R16B, FGF12, FEZF2, GDF7, and KCNQ5 in cancer and matched control tissue.

REFERENCES

- [1] Esteller, M. (2008). Epigenetics in cancer. *New England Journal of Medicine*, 358(11), 1148-1159.
- [2] Jones, P. A., & Baylin, S. B. (2002). The fundamental role of epigenetic events in cancer. *Nature Reviews Genetics*, 3(6), 415-428.
- [3] Locke, W. J., Guanzon, D., Ma, C., Liew, Y. J., Duesing, K. R., Fung, K. Y., & Ross, J. P. (2019). DNA methylation cancer biomarkers: Translation to

the clinic. *Frontiers in Genetics*, 1150.

- [4] IARC (2020). Globocan-2020. Retrieved from:
<https://gco.iarc.fr/today/data/factsheets/populations/356-india-factsheets.pdf>
- [5] Mathur, P., Sathishkumar, K., Chaturvedi, M., Das, P., Sudarshan, K. L., Santhappan, S., Nallasamy, V., John, A., Narasimhan, S., Roselind, F. S., & others. (2020). **Cancer** statistics, 2020: Report from national **cancer** registry programme, India. *JCO Global Oncology*, 6, 1063–1075.
- [6] Shanker, N., Mathur, P., Das, P., Sathishkumar, K., Shalini, A. M., & Chaturvedi, M. (2021). **Cancer** scenario in North-East India & need for an appropriate research agenda. *The Indian Journal of Medical Research*, 154(1), 27.
- [7] Lamare, F., Khongsti, S., Marthong, L., Ghosh, S., Chenkual, S., & Dkhar, H. et al. (2022). Genome-wide DNA methylation profiling of stomach **cancer** in the ethnic population of Mizoram, North East India. *Genomics*, 114(5), 110478.

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